

Module 2

Sustainability and Systems Thinking in Engineering



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Content

- Introduction to Sustainability and Systems Thinking
- Role of Engineers in Driving Sustainable Development
- Sustainable Engineering Tools and Approaches
- Energy and Material Sustainability in Engineering
- Sustainable Design Principles
- Practices and Case Studies



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Module Introduction

- Understand how engineering decisions impact the environment, economy, and society.
- Teaches the role of engineers in addressing global challenges like climate change, pollution, and resource depletion.
- Explore tools like life cycle assessment and carbon footprint analysis to design responsibly.
- Study sustainable use of energy and materials in engineering practices.
- Analyze real-world case studies to see how sustainability is implemented in engineering solutions.



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Why Sustainability?

- Irreparable damage caused to the environment by
 - Industrialization
 - Consumerism
- **Industrialization** is when people start making goods with machines in factories instead of by hand.
- **Consumerism** is the habit of buying and using more goods and services than are necessary.

SUSTAINABILITY

- Comes from the word “Sustain”
- Sustain : prolong for future



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Why Sustainability?

- Industrial Civilization and Consumerism – **Advantage**

- Economic growth
- Technological advancement
- Job creation
- Improved infrastructure
- Higher living standard

- Industrial Civilization and Consumerism – **Disadvantage**

- Environmental Degradation
- Waste generation
- Resource depletion
- Climate change
- Urban overcrowding



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Sustainability

Sustainability means **meeting our needs today** (like food, water, shelter, and energy) **without compromising the ability of future generations** to meet their own needs - Brundtland Commission (1987)

A sustainable solution must balance all three — Environment, Society, and Economy



3 P's

- ☐ **Planet**
- ☐ **People**
- ☐ **Profit**

Key Concepts

United Nations presented the following key sustainability concepts

1. Inter-generational equity
2. Intra-generational equity

Intergenerational equity: Fairness between different generations – ensuring future generations have the same opportunities, resources, and environmental quality as the present generation.

(Example: Conserving forests so your grandchildren can benefit from them too.)

Intragenerational equity: Fairness within the same generation – ensuring people living today share resources and opportunities equitably.

(Example: Reducing poverty and inequality among people living now.)



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Triple Bottom Line Solutions (3 Pillars of Sustainability)

Environmental Sustainability

Focuses on preserving the planet's natural resources and ecosystems for current and future generations by minimizing environmental degradation.

- Objectives
 - Reduce pollution and greenhouse gas emissions.
 - Conserve biodiversity and protect ecosystems.
 - Transition to renewable energy and sustainable resource



Triple Bottom Line Solutions (3 Pillars of Sustainability)

Social Sustainability

Ensures that the needs of people are met fairly and inclusively, promoting well-being, equality, and justice within and across communities.

- Objectives
 - Strengthen social cohesion and equity.
 - Foster inclusivity and access to basic services like education, healthcare, and housing.
 - Protect human rights and promote cultural diversity.



Triple Bottom Line Solutions (3 Pillars of Sustainability)

Economic Sustainability

Ensures that economic activities contribute to long-term growth without depleting natural resources or harming social well-being.

- Objectives
 - Create a stable economy that benefits all sections of society.
 - Encourage businesses to adopt sustainable practices.
 - Support innovation and efficient resource use.



ECONOMIC-SOCIAL AND ENVIRONMENTAL MATRIX

Quality of Life Concerns	Economic Issue		Social Issue		Environmental Issue	
	Unsustainable	Sustainable	Unsustainable	Sustainable	Unsustainable	Sustainable
Water	High cost of drinking water	Drinking water availability at low cost	Access to drinking water denied to weaker section	Adequate water supply to all sections	High-level of pollution in lakes and rivers	Conservation of existing fresh water bodies
Food	High cost of food and use of fertilizers and pesticides in farming	Good food available at low cost	Access to good food denied to weaker section	Adequate access to good food to all sections	Overuse of fertilizers and pesticides pollute the environment. Deforestation - conversion of forestland to farms	Food is of nutritious quality and related diseases are lowered.
Energy	High cost and intermittent power supply	Electricity available at low cost	Overuse of energy by the rich society and inadequate energy distribution	Adequate energy available to all sections	Use of fossil fuels and pollution	Use of renewable resources.(solar, wind, biomass)

Sustainable Development Goals (SDGs)

- The **Sustainable Development Goals (SDGs)** are 17 global goals established by the **United Nations in 2015**.
- They are part of the **2030 Agenda for Sustainable Development**, which aims to create a **better, fairer, and more sustainable world** for all people and the planet.



SDGs

SDG No.	Goal	SDG No.	Goal
1	No Poverty	10	Reduced Inequalities
2	Zero Hunger	11	Sustainable Cities and Communities
3	Good Health and Well-being	12	Responsible Consumption and Production
4	Quality Education	13	Climate Action
5	Gender Equality	14	Life Below Water
6	Clean Water and Sanitation	15	Life on Land
7	Affordable and Clean Energy	16	Peace, Justice and Strong Institutions
8	Decent Work and Economic Growth	17	Partnerships for the Goals
9	Industry, Innovation, and Infrastructure		



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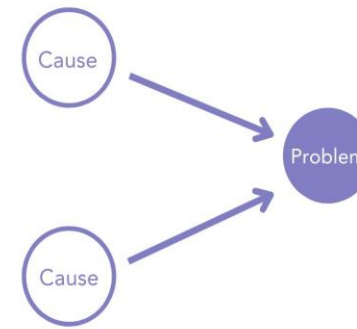
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System Thinking

- **Systems Thinking** is a way of understanding and solving problems by looking at the **whole system** rather than focusing on individual parts.
- It means thinking about how things are **connected**, how **one change affects another**, and how everything works together over time.



Linear Thinking

Reduce down to the most immediate and largest causes of the issue



Systems Thinking

Look at the network of factors that are creating the system's dynamics out of which the issue emerges

Key Features of System Thinking

- **Holistic Approach:** Problems are viewed in the context of the entire system.
- **Interconnectivity:** Focus on relationships and interactions between parts of the system.
- **Feedback Loops:** Recognition of feedback loops that can amplify or dampen effects within the system.
- **Emergence:** Understanding that the system as a whole exhibits behaviors and patterns that cannot be predicted by analyzing its parts in isolation.



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System Thinking and Sustainability

- Systems thinking helps us solve sustainability problems by looking at the entire system—how **environmental, social, and economic parts are connected**—rather than fixing one issue at a time.
- Sometimes, when we fix one problem without looking at the whole system, our solution accidentally creates a new problem somewhere else.
- Linear thinking: “**Plastic waste is bad → ban plastic bags.**”
- Systems thinking: “Why do people use plastic? → Low cost, lack of alternatives → solution: **promote reusable bags, redesign packaging, educate consumers, and adjust policies together.**”



Steps in Systems Thinking

- **Understand the real problem** – Don't jump to quick fixes. Ask “What's really happening?”
- **Draw the whole picture** – List all stakeholders and resources involved. Show how energy, materials, or information flow.
- **See how things are connected** – Find causes and effects. Look for loops where one change leads back to affect the system again.
- **Find smart points to act** – Look for places where a small change can make a big difference.
- **Think ahead about side effects** – Ask “If we do this, what else could happen later?”
- **Try it out and improve** – Test ideas on a small scale or in a model. Fix and refine before going big.



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Difference Between Linear and System Thinking

Aspect	Linear Thinking	Systems Thinking
Viewpoint	Looks at individual parts in isolation	Looks at the whole system and its connections
Cause and Effect	One cause → one effect (straight line)	Multiple causes → interconnected effects (loops)
Problem-Solving	Fixes visible symptoms quickly	Addresses root causes for lasting solutions
Time Horizon	Short-term focus	Long-term impact considered
Consequences	Often creates new problems elsewhere	Prevents unintended side effects
Approach	Add more resources to meet demand	Reduce waste, optimize, and redesign the system



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Example of System Thinking

Problem: A website is slow.

- **Linear Thinking:** Increase server speed.
- **Systems Thinking:**
 - Check if images are too large
 - If internet speed is affecting users
 - If database queries are slow
 - If too many people are visiting the site at once
- **Solution:** Optimize the website first — compress images, remove unnecessary code, and use caching to make pages load faster.



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Example of System Thinking

Problem: A road has traffic jams every morning.

- **Linear Thinking:** Widen the road.
- **Systems Thinking:**
 - Observe if school timings and office hours overlap
 - Check for signal timings
 - Look at public transport availability and accessibility
 - Consider pedestrian crossings slowing traffic
- **Solution:** Adjust signals, stagger timings, promote bus travel—not just widen the road.



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Example of System Thinking

Problem: Electricity demand is rising rapidly.

- **Linear Thinking:** Build more coal power plants.
- **Systems Thinking:**
 - Check if industries and homes are wasting power (inefficient appliances, poor insulation)
 - Look at whether renewable energy sources (solar, wind) are being fully used
 - Study if people are aware of energy-saving practices or have incentives to save
- **Solution:** Improve energy efficiency, add renewable power, encourage off-peak usage and conservation—not just build more coal plants

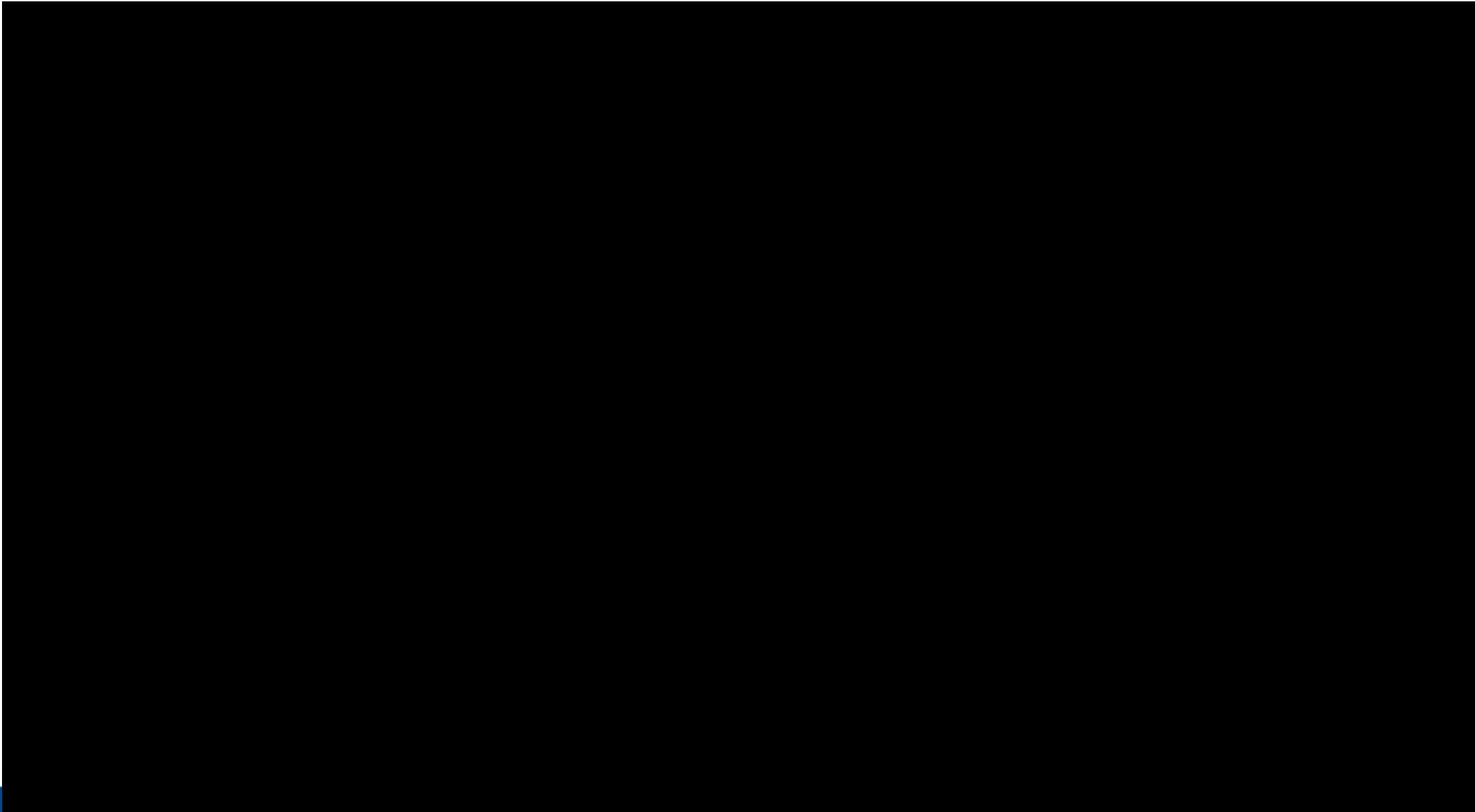


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Why Do Engineers Need to Learn Sustainability?

As engineers:

- You will **design infrastructure and systems** that use water, energy, and materials.
- Your decisions will **impact energy use, pollution, and waste generation**.
- You will help **solve real-world problems** that affect people, communities, and the environment.
- Engineers are **change-makers**. You must ensure that what you build is:
 - **Safe** for people
 - **Inclusive** for all
 - And **responsible** toward the environment.



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Role of Engineers in Driving Sustainable Development

Creating Solutions that Protect the Planet:

- Engineers design systems that reduce pollution, conserve resources, and protect nature.

Civil: Design of green buildings using less energy and water

Computer Science: Software to monitor energy usage in buildings

Improving Efficiency in Every Field

- Engineers work to optimize systems—making them smarter, faster, and cleaner.

Mechanical : Fuel-efficient engines

Computer Science : Algorithms that reduce server load and power usage in data centers



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Role of Engineers in Driving Sustainable Development

Building for People and Future Generations:

- Engineering solutions should be inclusive, long-lasting, and safe for all.

Electrical: Solar-powered lighting for rural areas

Computer Science: Assistive tech apps for visually or hearing-impaired users

Solving Global Challenges Locally:

- Engineers help solve issues like climate change, clean water, and urban pollution through localized innovation.

Environmental: Water filters for rural households

Computer Science: IoT-based waste bin monitoring to reduce overflow in cities



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Role of Engineers in Driving Sustainable Development

Supporting the UN Sustainable Development Goals (SDGs):

- Engineers contribute to many SDGs, including clean water (SDG 6), clean energy (SDG 7), sustainable cities (SDG 11), industry ,innovation and infrastructure (SDG 9) and climate action (SDG 13).



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How Engineers Should Adapt Their Approach for Sustainability

1. **From Quantity to Quality:**

Focus has shifted from producing more resources to ensuring their quality, safety, and minimal environmental impact.

2. **From Centralized to Mixed Systems:**

Moving from only large-scale centralized solutions to a mix of centralized and localized strategies for flexibility and resilience.

3. **From Linear to Circular Models:**

Transitioning from “take–make–dispose” models to **circular systems** that prioritize reuse, recycling, and resource recovery.



How Engineers Should Adapt Their Approach for Sustainability

4. **From Government-Only to Collaborative Governance:**
Shifting from government-driven actions to **participatory approaches** involving industries, communities, and individuals.
5. **From Reactive to Proactive Planning:**
Evolving from responding to crises after they happen to **anticipating risks** and designing for resilience, especially against climate change.



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Challenges in implementing Sustainability

1. **High costs** – Initial investments in sustainable practices can be expensive.
2. **Lack of awareness** – People and organizations often don't fully understand sustainability.
3. **Policy gaps** – Inconsistent or weak regulations hinder progress.
4. **Technological limitations** – Some required green technologies are still emerging.
5. **Resistance to change** – Industries and communities may prefer conventional methods.
6. **Limited resources** – Funding, expertise, or materials may not be available.



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Challenges in implementing Sustainability

7. **Measurement difficulties** – Tracking sustainability impacts can be complex.
8. **Conflicting priorities** – Economic growth goals may clash with environmental protection.
9. **Social inequality** – Unequal access to benefits slows adoption in some regions.
10. **Global coordination issues** – Countries differ in commitment and capability.



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Quiz for Understanding

You are designing a new college building. Which option is the most sustainable feature?

- a. Installing centralized air-conditioning for comfort
- b. Using large glass panels for natural light and aesthetics
- c. Installing rooftop solar panels and a rainwater harvesting system
- d. Painting the building white to reflect heat

Ans: c. Installing rooftop solar panels and a rainwater harvesting system



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Quiz for Understanding

Your city faces traffic congestion and air pollution. Which solution is the most sustainable?

- a. Building more flyovers
- b. Promoting Car Ownership
- c. Widening of roads after cutting trees
- d. Developing a public transport app with real-time bus tracking

Ans: d. Developing a public transport app with real-time bus tracking



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Module 2

Sustainability and Systems Thinking in Engineering



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Sustainable Engineering Tools and Approaches

- Sustainable Engineering Tools and Approaches are structured **methods, frameworks, and techniques** that engineers use to **design, evaluate, and improve systems, products, and processes** in ways that minimize **environmental harm, support social well-being, and ensure economic viability** over the long term.
- They help engineers **make informed, data-driven decisions** to reduce **resource use, lower emissions, improve efficiency, and align projects with sustainability goals** such as the **UN SDGs**.



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Sustainable Engineering Tools and Approaches

- Life Cycle Assessment (LCA)
- Carbon Footprint
- Embodied Energy



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Life Cycle Assessment (LCA)

- **Life Cycle Assessment is a systematic method to evaluate the environmental impacts of a product, process, or service throughout its life cycle — from raw material extraction to disposal ("cradle to grave").**
- Why it matters for engineers:
- Engineers design products, processes, and infrastructure. Decisions made during design affect:
 - Energy consumption
 - Material usage
 - Waste generation
 - Pollution



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Life Cycle Stages

(Example of Paper Manufacturing)

1. Raw Material Extraction

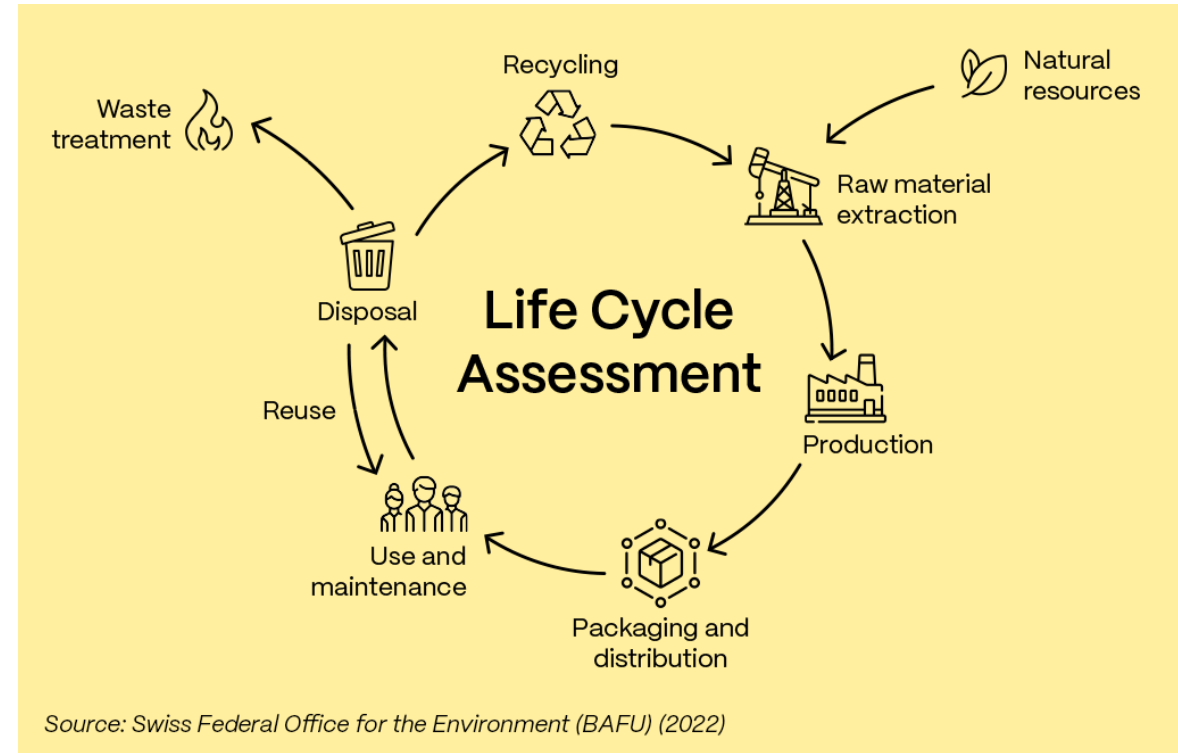
- Logging of trees, water usage, transport of raw wood
- **Environmental impacts: deforestation, biodiversity loss, fuel emissions**

2. Material Processing

- Wood pulping, bleaching, and chemical treatment
- **Environmental Impacts: chemical waste, water pollution, high energy consumption**

3. Manufacturing

- Paper sheet formation, drying, cutting, packaging
- **Environmental Impacts: CO₂ emissions, solid waste**



Life Cycle Stages

(Example of Paper Manufacturing)

4. Distribution

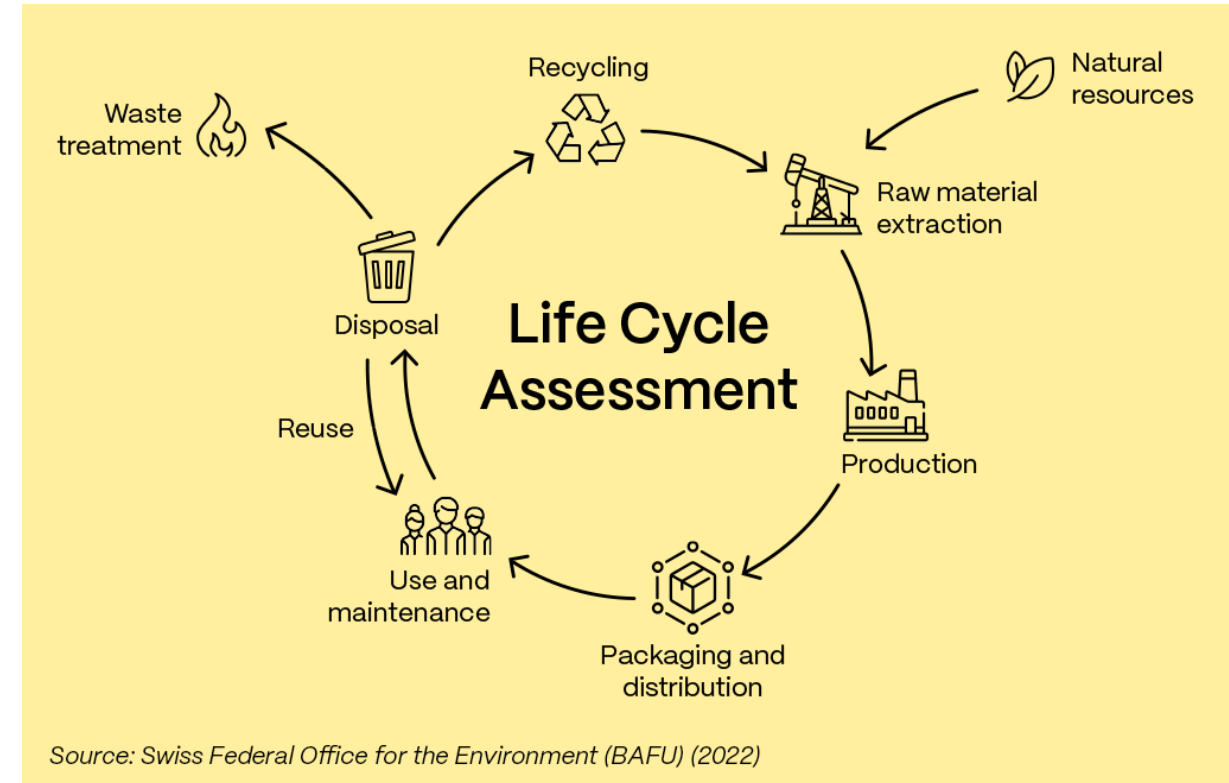
- Transport to wholesalers, retailers, customers
- **Environmental Impacts: fuel consumption, emissions**

5. Use Phase

- Writing, printing

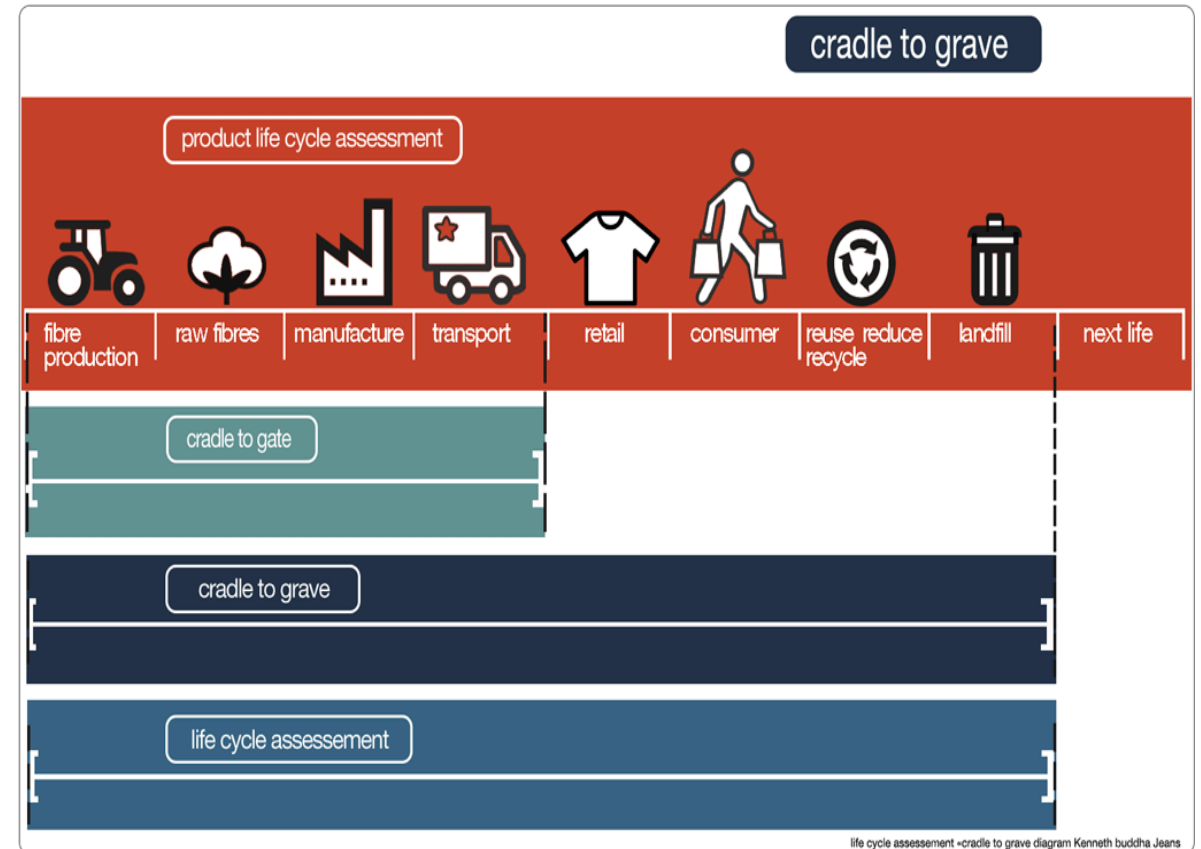
6. End-of-Life

- Landfilling, incineration, or recycling
- **Environmental Impacts: methane emissions from landfill, ash from incineration, or benefits from recycling**



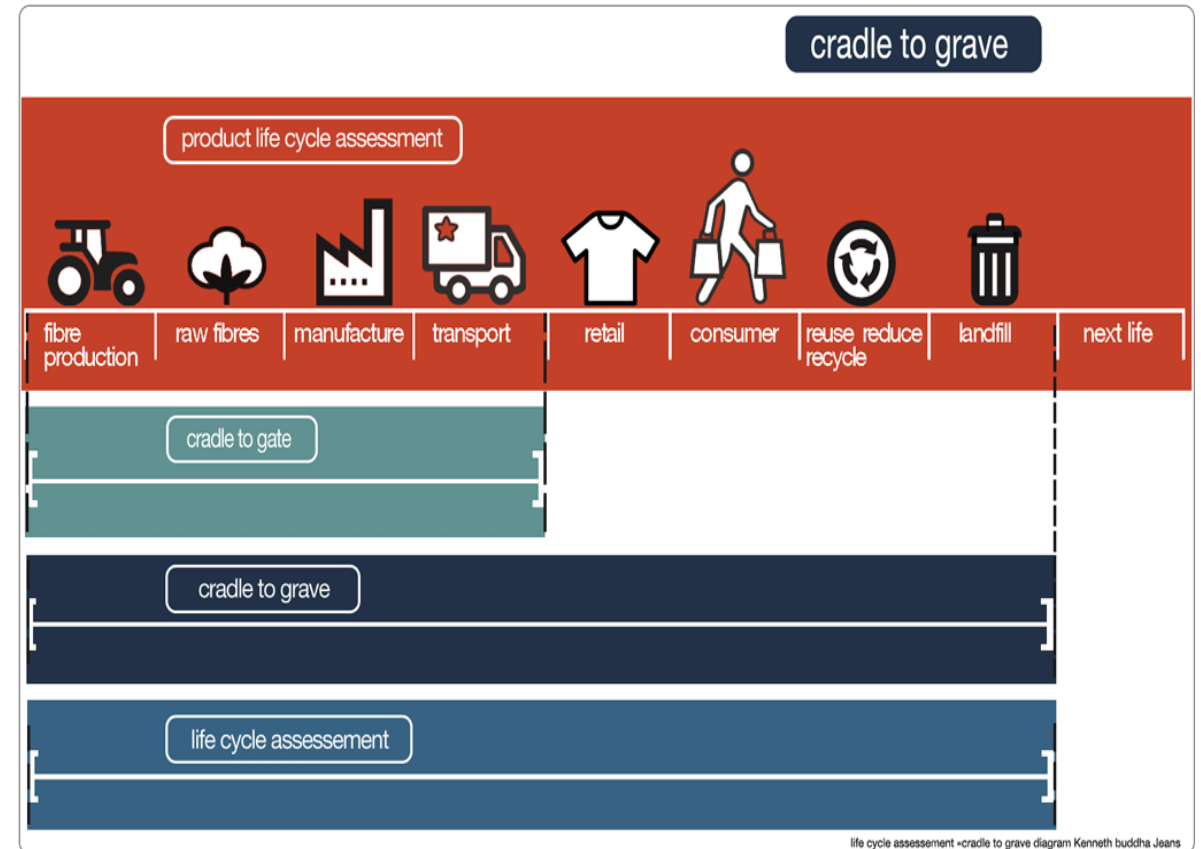
Types of LCA

- **Cradle to Grave** - Considers the entire life cycle of a product, **from raw material extraction to the end-of-life stage, including disposal**. This provides a comprehensive view of the product's environmental impact.
- **Cradle to Gate** - Examines the environmental impact from the **extraction of raw materials to the point where the product leaves the manufacturing facility**. This approach provides insights into the initial stages of a product's life cycle.



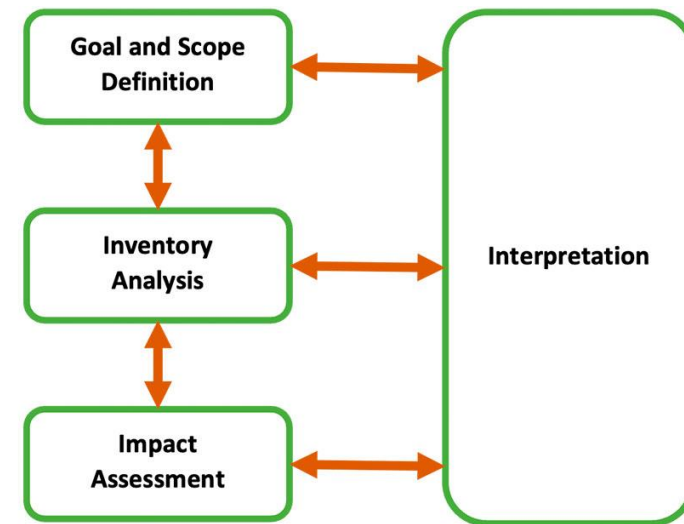
Types of LCA

- **Gate to Gate-** Focuses on the environmental impact of a **specific process within a product's manufacturing chain, from the entry point to the exit point.** It doesn't consider raw material extraction or product use and disposal.



LCA Frame Work

- The Life Cycle Assessment (LCA) framework is the **structured, step-by-step approach**—outlined in the **International Organization of standardization (ISO 14040 and 14044 standards)**—that guides how an LCA is conducted, from defining the purpose to interpreting the results.
- It is conducted in four stages
 - Goal and Scope Definition
 - Life Cycle Inventory (LCI)
 - Life Cycle Impact Assessment (LCIA)
 - Interpretation



Step 1 – Goal and Scope Definition



- Define **purpose** - why you are doing the LCA
- Define **functional unit** - A measurable reference point for comparison
- Define **system boundaries** - what processes are included/excluded

Example:

- **Goal:** Assess the environmental impact of producing one cotton T-shirt.
- **Functional Unit:** 1 T-shirt (~150 g cotton).
- **System Boundaries:** Cradle-to-Grave (cotton farming → fabric production → manufacturing → distribution → use → end-of-life).

Step 2 – Life Cycle Inventory (LCI)



This is the **data collection phase** of all inputs and outputs.

- Energy inputs
- Raw material inputs
- Emissions to air, water, and land
- Waste generation

• Example:

Inputs

- Cotton: 0.15 kg
- Water: ~2700 liters (cotton farming & processing)
- Energy: 3 kWh for manufacturing
- Chemicals: dyes, detergents

Outputs

- CO₂ emissions: ~7 kg CO₂-eq over lifetime
- Wastewater: dye and detergent residues
- Solid waste: fabric offcuts (~20 g), end-of-life disposal waste



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Step 3 – Life Cycle Impact Assessment (LCIA)

- Translate inventory data into environmental impact indicators (e.g., CO₂ emissions, water footprint)
- Converting raw data into environmental impact categories.

Example:

- **Global Warming Potential:** High from electricity for washing/drying.
- **Water Consumption:** Largest impact during cotton farming (irrigation).
- **Chemical Pollution:** Dyes during fabric processing; detergents during washing.



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Step 4 – Interpretation

- Analyze results, identify hotspots, and recommend improvements.
- The Interpretation phase is not a one-time final step; it's an ongoing process that happens throughout the entire LCA.

Example:

- **Hotspots:**
 - Cotton cultivation (water-intensive).
 - Use phase (washing and drying) consumes most energy.
- **Improvement Options:**
 - Promote organic cotton or rain-fed cotton to reduce water use.
 - Extend garment life by reducing washing frequency.



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Benefits of LCA

- Helps design sustainable products
- Reduces resource use and emissions
- Complies with environmental regulations
- Improves brand image and competitiveness
- Informs policy and procurement decisions

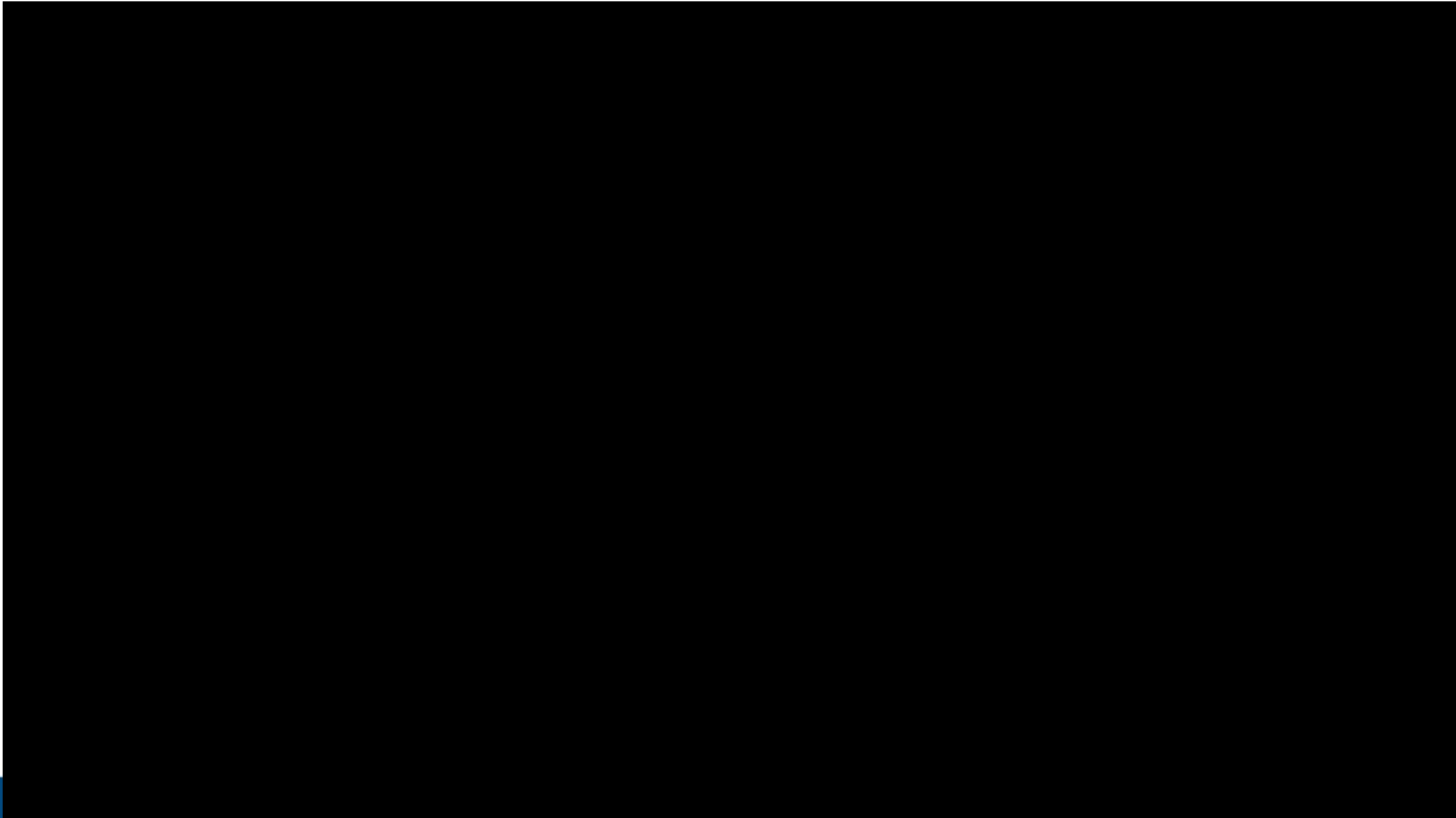


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Module 2

Sustainability and Systems Thinking in Engineering



Topics on Carbon Footprint

- Introduction to Carbon Footprint
- Sources of Carbon Footprint
- Measuring Carbon Footprint
- Reducing Carbon Footprint – Sustainable Practices
- Problems on carbon footprint



Carbon Footprint

- Carbon footprint is the **total amount of greenhouse gases (GHGs) released into the atmosphere because of an activity, product, person, or organization — expressed as carbon dioxide equivalent (CO₂-eq).**
- Reducing the carbon footprint is critical for mitigating climate change and transitioning to a sustainable future

Green House Gases (GHGs)

- Carbon dioxide (CO₂): burning fossil fuels, deforestation
- Methane (CH₄): agriculture (cattle, rice fields), landfills, oil & gas leaks
- Nitrous oxide (N₂O): fertilizers, industrial emissions
- Fluorinated gases (HFCs, PFCs, SF₆, NF₃): refrigerants, electronics manufacturing

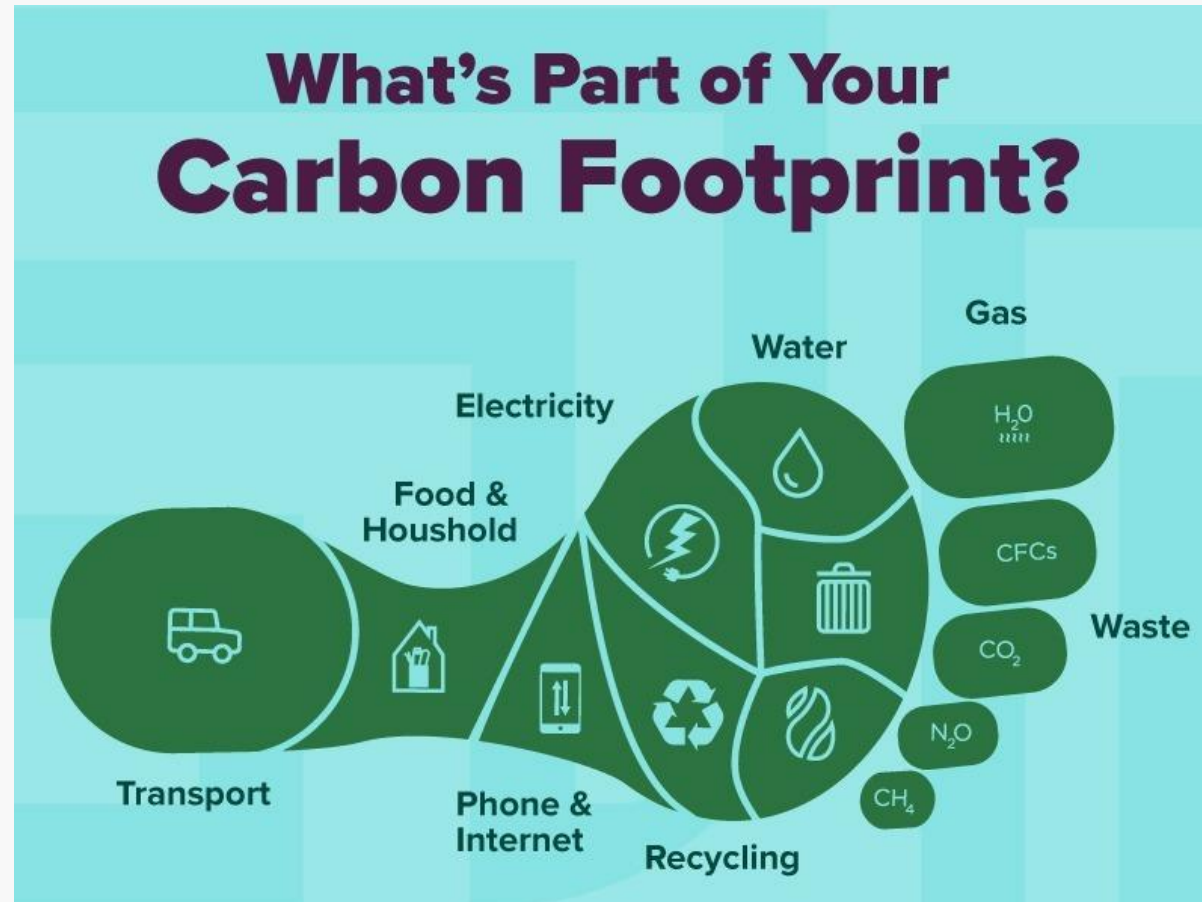


Examples :

Charging phones/laptops → electricity → CO₂ from coal power plants.

Travelling by petrol/diesel buses.

Eating packaged food (transport + plastic packaging).



Sources of Carbon Footprint

- **Travel** (cars, flights, buses).
- **Energy** (fans, AC, lights, computers).
- **Food** (meat has 4× higher footprint than vegetables).
- **Gadgets** (manufacturing laptops/phones uses metals, plastic, electricity).
- **Data Centers**: run 24/7; need cooling → consume huge energy.
- **AI/Cloud Computing**: running models = high power demand.
- **E-Waste**: old mobiles, laptops → toxic metals pollute landfills.
- **Internet**: streaming 1 hour of HD video $\approx$ 55 g CO₂ (electricity).
- **Industries**: steel, cement = high CO₂ emission.
- **Transport sector**: trucks, trains, planes → burning fossil fuels.
- **Construction**: concrete → produces CO₂ when making cement.



Measuring Carbon Footprint

Carbon footprint is measured in CO₂ equivalent (CO₂e).

- 1. Direct measurements:** Emissions from **power consumption, transportation, manufacturing processes.**
- 2. Indirect measurements:** Emissions embedded in products/services, supply chain.

Tools: Online calculators, GHG Protocol, life cycle assessment (LCA) tools.

Use: Companies assess the footprint to improve efficiency and report sustainability.



Reducing Carbon Footprint – Sustainable Practices

1. **Green computing & energy-efficient code** – Writing optimized programs that use less processing power and memory reduces the overall electricity consumption of computers and servers.
2. **Cloud servers on renewable energy** – Many companies are shifting their data centers to run on solar, wind, or hydro energy instead of coal-based power, making digital services more sustainable.
3. **Digital hygiene** – Simple habits like deleting unused files, unsubscribing from unnecessary emails, and streaming in standard quality instead of HD can significantly cut down the energy load of internet usage.
4. **Renewable energy (solar, wind, hydro)** – Replacing fossil fuels with renewable energy sources can drastically reduce CO₂ emissions while ensuring reliable power for industries and communities.



Reducing Carbon Footprint – Sustainable Practices

- 5. Green buildings & sustainable materials** – Designing buildings that use natural lighting, better insulation, and eco-friendly materials reduces both construction-related emissions and long-term energy use.
- 6. Public transport & electric mobility** – Encouraging metro, buses, carpools, and electric vehicles instead of personal petrol/diesel cars reduces urban air pollution and carbon emissions.
- 7. Cycle/walk for short trips** – Choosing cycling or walking instead of motorbikes or cars for nearby travel saves fuel and promotes health while cutting emissions.
- 8. Switch off unused devices** – Turning off lights, fans, laptops, and mobile chargers when not in use prevents energy wastage and lowers electricity bills.
- 9.Reduce plastic & e-waste** – Avoiding single-use plastics and properly recycling old mobiles or laptops prevents harmful waste from entering landfills and reduces the carbon footprint of manufacturing new items.



Problems on carbon footprint

1) A student uses a laptop for 5 hours a day. The laptop consumes 50 watts per hour. The emission factor for electricity is 0.8 kg CO₂/kWh. Calculate the **student's weekly carbon footprint** from laptop usage in kg CO₂.

Solution:

- Daily use = 5 hours
- Power consumption = 50 W = 0.05 kW
- Weekly use = 5 hours/day × 7 days = 35 hours
- Energy consumed = 0.05 kW × 35 hours = 1.75 kWh
- Carbon footprint = 1.75 kWh × 0.8 kg CO₂/kWh = ***1.4 kg CO₂ per week***



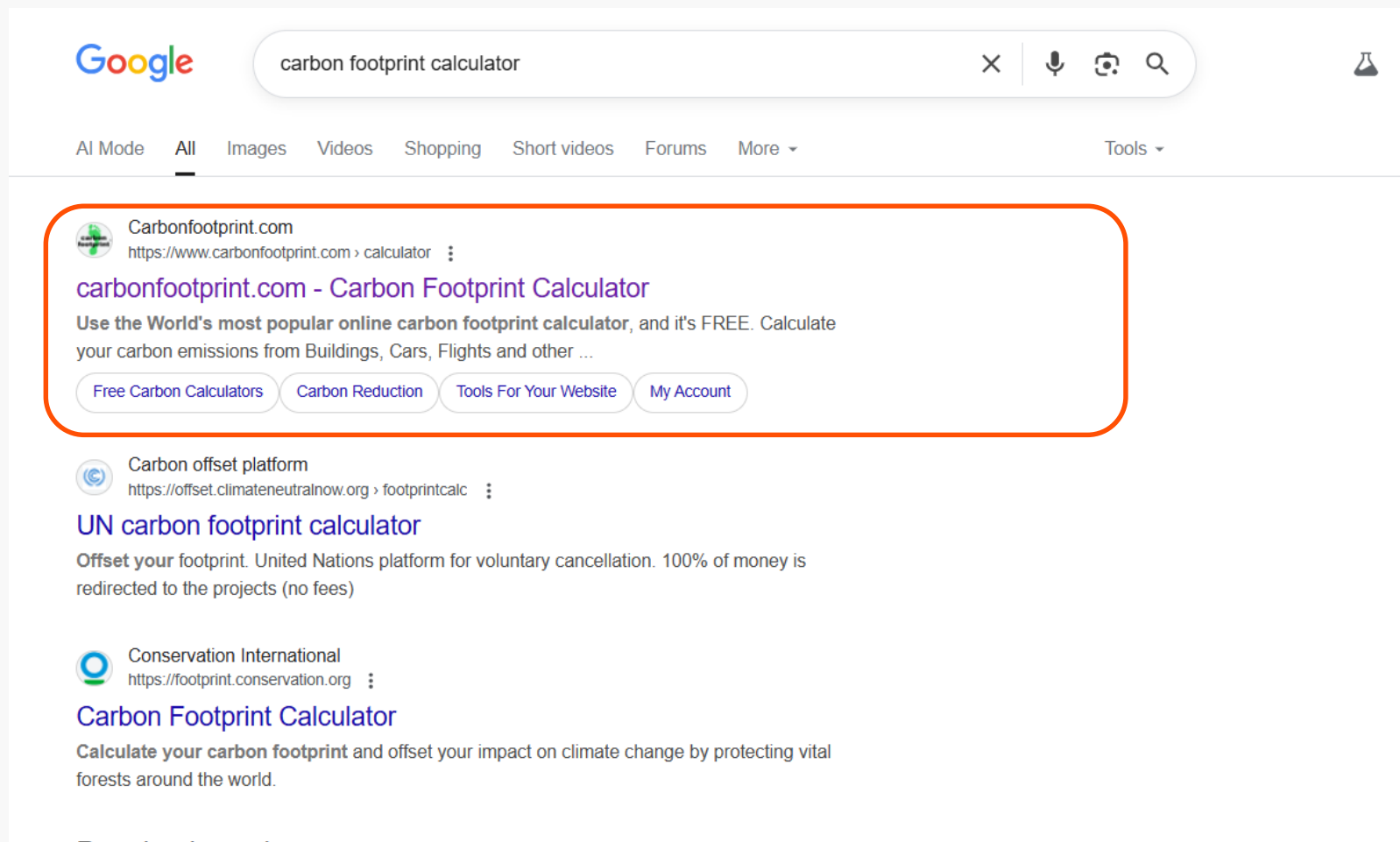
2) computer lab uses three air conditioners, each rated 1.5 kW, for 8 hours a day, 25 days a month. The emission factor for electricity is 0.9 kg CO₂/kWh. **Calculate the monthly carbon footprint from these AC units.**

Solution:

- Total AC power = $3 \times 1.5 \text{ kW} = 4.5 \text{ kW}$
- Hours/month = $8 \times 25 = 200 \text{ hours}$
- Total energy = $4.5 \text{ kW} \times 200 \text{ h} = 900 \text{ kWh}$
- Emissions = $900 \text{ kWh} \times 0.9 \text{ kg CO}_2/\text{kWh} = \textbf{810 kg CO}_2 \textbf{ per month}$



Carbon footprint calculator



<https://www.carbonfootprint.com/calculator.aspx?utm>

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Module 2

Sustainability and Systems Thinking in Engineering



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Sustainable Engineering Tools and Approaches

- Life Cycle Assessment (LCA)
- Carbon Footprint
- Embodied Energy



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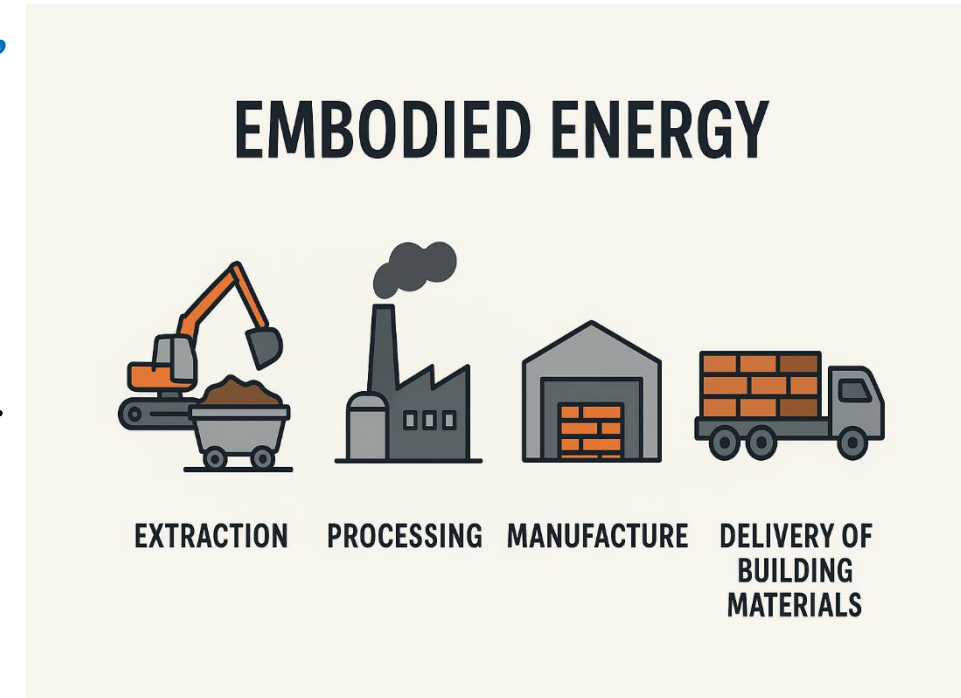
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Embodied Energy

- **The total energy consumed to produce a material, component, or product up to the point of use.**
- This includes all the energy required for:
 - **Extraction** – Mining, logging, or collecting raw materials.
 - **Processing** – Converting raw materials into usable form.
 - **Manufacture** – Making building products (like cement, steel rods).
 - **Delivery** – Transporting to construction site.
- It does **not** include the energy consumed during the (i) product's operation and maintenance; (ii) disposal or recycling



Embodied Energy

- High embodied energy means the product has already caused significant environmental impact before even being used.
- In sustainable engineering and green building, reducing embodied energy helps lower overall carbon emissions and resource depletion.
- For example, aluminum has very high embodied energy because smelting requires a lot of electricity, while timber usually has low embodied energy since it needs minimal processing.
- **Lower embodied energy = less environmental impact.**
- **Embodied Energy helps in choosing sustainable building materials.**
- **Units: MJ/kg (megajoules per kilogram) or MJ/m² (megajoules per area).**



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Application of Embodied Energy

- Manufacturing industry: Cars, airplanes, and ships — energy used to extract metals, make components, and assemble.
- Consumer goods: Electronics, furniture, textiles, packaging — energy embedded in raw material processing and production.
- Food industry: Energy required to grow crops, process food, package, and transport it (often called embodied energy of food).
- **Infrastructure projects:** Roads, bridges, dams — energy to produce asphalt, steel, and concrete.
- **Buildings consume massive material quantities, and construction materials like cement, steel, and aluminum have very high embodied energy.**



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Factors affecting Embodied Energy

- **Material type:** Metals (like aluminum, steel) usually have high embodied energy; natural materials (like timber) are lower.
- **Manufacturing technology:** Efficient or renewable-energy-powered processes reduce embodied energy.
- **Recycled content:** Using recycled materials generally lowers embodied energy compared to virgin material production.
- **Location and transport distance:** Imported materials have higher transport energy.
- **Construction technique:** Prefabricated components may reduce on-site energy but could increase factory energy consumption.



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Embodied Energy of various material

- **Aluminum: 150–250 MJ/kg**
- **Steel: 20–35 MJ/kg**
- **Cement: 4–5 MJ/kg**
- **Concrete: 0.8–1.5 MJ/kg**
- **Glass: 12–15 MJ/kg**
- **Timber: 2–3 MJ/kg**



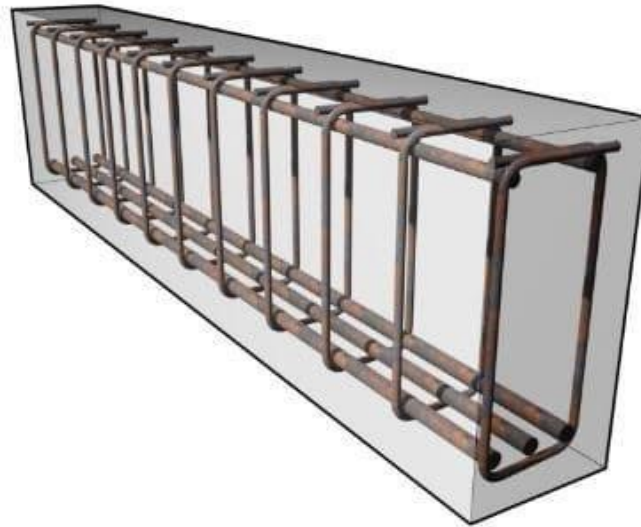
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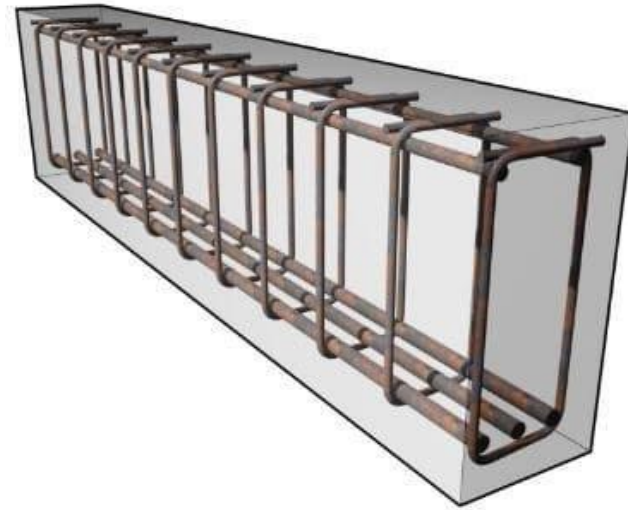
**Calculate the total embodied energy
for the Reinforced Cement Concrete
beam structure of volume 5 m³**



- **Step I –
Total Volume of Beam
= Volume of Cement Concrete + Volume of steel**

**Calculate the total embodied energy
for the Reinforced Cement Concrete
beam structure of volume 5 m³**

- Let us assume the Percentage of steel to be 1.5 %
- Steel Volume = 1.5 % (Total Volume)
- Steel Volume = $(1.5 \times 5) / 100$
- **Steel Volume = 0.075 m³**
- **Volume of concrete**
- **= 5 - 0.075 = 4.925 m³**



Mass calculation

- Density of steel = 7850 kg/m^3
- Density of Concrete = 2400 kg/m^3

Density = Mass/Volume

- Mass = Density *Volume
- Mass of Steel = $7850 * 0.075 = 588.75 \text{ Kg}$
- Mass of Concrete = $2400 * 4.925 = 11820 \text{ Kg}$

Embodied Energy

- ✓ Concrete - 0.8 MJ/Kg
- ✓ Steel - 20 MJ/Kg
- Embodied Energy = Mass * (Embodied energy per mass)
- Embodied Energy of steel = $588.75 * 20 = 11775 \text{ MJ}$
- Embodied Energy of concrete = $11820 * 0.8 = 9456 \text{ MJ}$
- **Total Embodied energy =**
Embodied energy of steel + Embodied energy of concrete
- **Total Embodied Energy = $11775 + 9456 = 21231 \text{ MJ}$**

Problem 2

An RCC beam has a **total volume = 7.0 m^3** . The design uses steel reinforcement equal to **1.5%** of the beam volume.

Given:

Density of steel = 7850 kg/m^3

Density of concrete = 2400 kg/m^3

Embodied energy of steel = 8 MJ/kg

Embodied energy of concrete = 0.8 MJ/kg

- Calculate the **total embodied energy** of the beam for the original design ((Fe415) (steel = 1.5% of volume).
- If higher-strength steel (Fe500) is used so that the required steel volume reduces to **1.1%** of the beam volume, calculate the **total embodied energy** for this Fe500 design.
- Compute the **percentage reduction in total embodied energy** of the Fe500 design compared to the original (1.5%) design. Which design produces the greater embodied-energy saving?



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(a) Original design — steel = 1.5% of volume

- Steel volume: $V_{s1} = 0.015 \times 7.0 = 0.105 \text{ m}^3$
- Concrete volume: $V_{c1} = 7.0 - 0.105 = 6.895 \text{ m}^3$
- Mass of steel: $m_{s1} = 7850 \times 0.105 = 824.25 \text{ kg}$
- Mass of concrete: $m_{c1} = 2400 \times 6.895 = 16\,548.00 \text{ kg}$
- EE (steel): $EE_{s1} = 824.25 \times 20.0 = 16\,485.00 \text{ MJ}$
- EE (concrete): $EE_{c1} = 16\,548.00 \times 0.8 = 13\,238.40 \text{ MJ}$

$$\text{Total EE (original)} = 16\,485.00 + 13\,238.40 = \boxed{29\,723.40 \text{ MJ}}$$

(b) Fe500 design — steel = 1.1% of volume

- Steel volume: $V_{s2} = 0.011 \times 7.0 = 0.077 \text{ m}^3$
- Concrete volume: $V_{c2} = 7.0 - 0.077 = 6.923 \text{ m}^3$
- Mass of steel: $m_{s2} = 7850 \times 0.077 = 604.45 \text{ kg}$
- Mass of concrete: $m_{c2} = 2400 \times 6.923 = 16\,615.20 \text{ kg}$
- EE (steel): $EE_{s2} = 604.45 \times 20.0 = 12\,089.00 \text{ MJ}$
- EE (concrete): $EE_{c2} = 16\,615.20 \times 0.8 = 13\,292.16 \text{ MJ}$

$$\text{Total EE (Fe500)} = 12\,089.00 + 13\,292.16 = \boxed{25\,381.16 \text{ MJ}}$$



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(c) Savings (Fe500 vs original)

- Absolute saving: $29\,723.40 - 25\,381.16 = 4\,342.24 \text{ MJ}$
- Percentage saving: $\frac{4\,342.24}{29\,723.40} \times 100 \approx 14.61\%$

With steel EE = **20 MJ/kg**, switching from 1.5% to 1.1% steel (Fe500) saves **~4,342 MJ**, i.e. **~14.6%** reduction in total embodied energy.



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Ways to Reduce Embodied Energy

1. Use **local materials** to reduce transport energy.
2. Select **low-energy, renewable, or natural materials** (e.g., timber, bamboo).
3. Increase **recycled content** in steel, aluminum, glass, etc.
4. Opt for **efficient manufacturing methods** and renewable-powered production.
5. Design for **durability and adaptability**, reducing recurring embodied energy.
6. Plan for **reuse and recycling at end of life** to close the loop.



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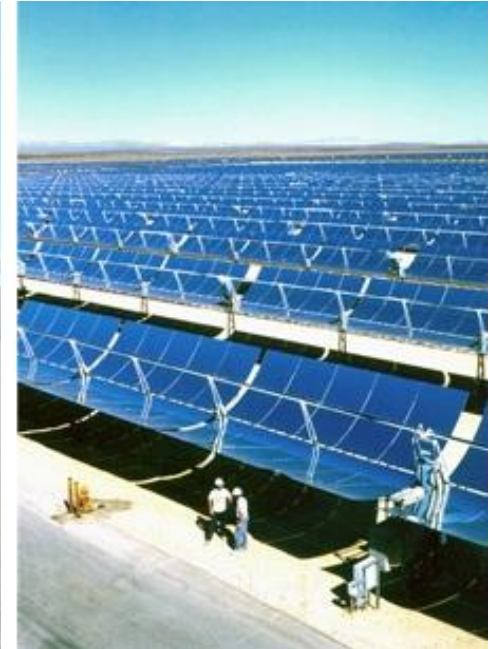
Module 2

Sustainability and Systems Thinking in Engineering



Energy Sustainability in Engineering

- Energy sustainability is the concept of meeting the **world's energy needs** today without compromising the ability of future generations to meet their own energy needs.
- It focuses on **producing, distributing, and consuming energy** in ways that are environmentally sound, economically viable, and socially responsible.
- **Global energy demand is increasing** → need for efficiency and shifting to renewable energy source should be considered.



Energy Sustainability in Engineering

Key Concepts

- Energy Efficiency
- Renewable Energy Source
- Decarbonization
- Energy Recovery



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Energy efficiency

- Energy efficiency means using less energy to perform the same task or achieve the same outcome.
- Instead of wasting energy, it ensures maximum output from minimum input.
- Key Idea: “Do more with less energy.”
- Benefits:
- Reduces energy bills.
- Lowers greenhouse gas emissions.
- Helps conserve natural resources.



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Energy efficiency

- Examples:
- LED bulbs provide the same amount of light as incandescent bulbs but consume 70–80% less electricity.
- A well-insulated building requires less heating or cooling energy.
- Modern appliances like 5-star rated refrigerators and ACs reduce electricity consumption.



Renewable Energy Source

- Renewable energy comes from sources that are naturally replenished and will not run out on a human timescale.
- Benefits:
- Clean and sustainable.
- Reduces dependence on fossil fuels like coal and oil.
- Plays a critical role in achieving global climate goals (SDG 7 & SDG 13).



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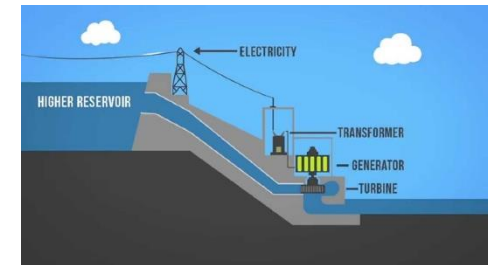
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Renewable Energy Source

- Types of Renewable Sources:
- Solar Energy – converting sunlight into electricity (solar panels).
- Wind Energy – wind turbines harnessing wind power.
- Hydropower – using flowing water to generate electricity.
- Biomass – using organic matter (crop residues, wood, waste) for energy.
- Geothermal Energy – using heat from the Earth's core.



Decarbonization

- Decarbonization means reducing carbon dioxide (CO₂) and other greenhouse gas (GHG) emissions that come mainly from burning fossil fuels.
- Why Important?: Fossil fuels like coal, oil, and natural gas are the primary contributors to global warming and climate change.
- Decarbonization is done to Achieve **Net Zero**
- Net Zero means balancing the amount of greenhouse gases (GHGs) emitted into the atmosphere with the amount removed from it.
- In other words → Emissions in = Removals out.



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Decarbonization

- Strategies for Decarbonization:

1. carbon capture and storage (CCS) technologies.

- Carbon Capture and Storage (CCS) is a technology to capture carbon dioxide (CO₂) emissions from large sources (like power plants, steel, cement factories), transport it, and then store it underground so it does not enter the atmosphere and contribute to global warming.

2. Carbon offset

- Carbon offset = compensating for emissions in one place by reducing or removing emissions elsewhere.
- Example: A aviation company funds tree plantation to offset the emissions from its flights.
- It is part of decarbonization, especially for sectors where emissions are hard to eliminate (like aviation, shipping, cement).



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Energy Recovery

- Energy recovery is the process of capturing and reusing energy that would otherwise be wasted.
- Key Idea: “Don’t let energy go to waste—recover and use it again.”
- Benefits:
- Reduces reliance on fossil fuels.
- Minimizes environmental pollution.
- Efficient waste management.



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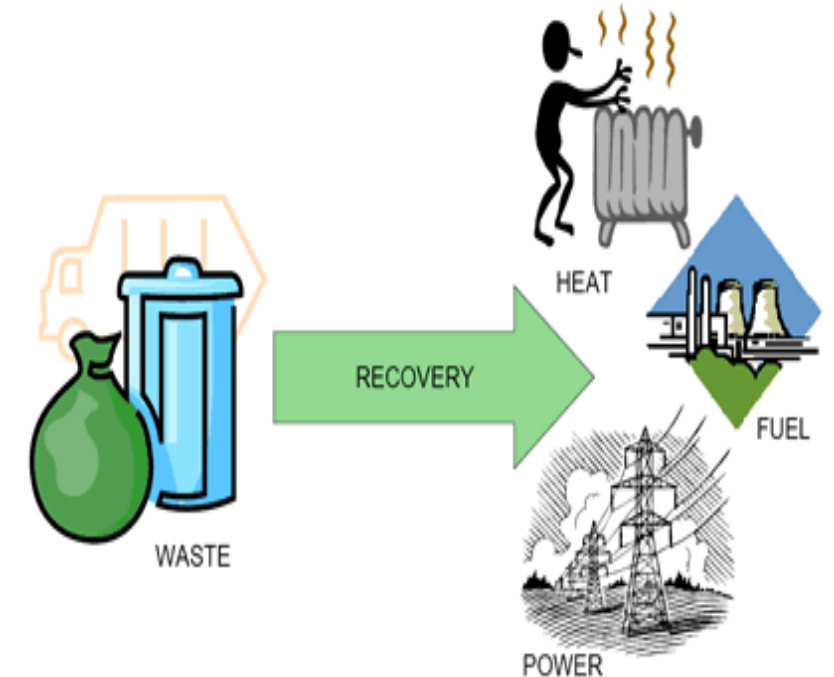
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Energy Recovery

- Examples:
- Waste-to-Energy (WTE) plants: Burning municipal solid waste to generate electricity.
- Heat Recovery Systems: Capturing waste heat from industries (like cement plants or steel mills) and using it for heating water or generating steam.
- Biogas from Waste: Organic waste (food, sewage, animal manure) converted into methane gas for cooking or electricity generation.



Energy Audit

- Energy Audit = Diagnosis tool.
- An Energy Audit is a systematic process of studying how energy is used in a building, factory, or system, identifying areas of energy wastage, and recommending measures to improve efficiency.
- Purpose:
 - To analyze energy consumption.
 - To reduce energy costs.
 - To minimize greenhouse gas emissions.
 - To create a roadmap for energy efficiency & sustainability.



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Energy Audit

Types of Energy Audits

1. Preliminary/Walk-through Audit

- Quick inspection → identifies obvious areas of energy waste.
- Example: Checking lighting, air conditioning, insulation, leaks.

2. Detailed Energy Audit

- In-depth measurement and analysis of energy use.
- Uses meters, sensors, software tools.

3. Investment-Grade Audit

- Most detailed → used for financial planning.
- Calculates return on investment (ROI) for energy-saving projects.



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Material Sustainability in Engineering

- Material sustainability in engineering refers to the **responsible selection, use, and management of materials** throughout their life to minimize environmental impacts, conserve resources, and support long-term sustainability.
- Benefits
- Reduces environmental footprint of engineering projects.
- Enhances resource efficiency and lowers costs over the life cycle.
- Supports SDG 9 (Industry, Innovation, Infrastructure) and SDG 12 (Responsible Consumption and Production).
- Contributes to decarbonization and net zero by lowering embodied emissions.



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Material Sustainability in Engineering

- Key Principles
- Efficient Use of Resources: Use fewer raw materials without compromising performance.
- Use of Renewable & Recycled Materials: Prioritize renewable (wood, bamboo) or recycled materials (recycled steel, plastic, fly ash in concrete).
- Durability & Longevity: Choose materials that last longer to reduce replacement needs.
- Circular Economy Approach: Design for reuse, recycling, and remanufacturing at end-of-life.
- Embodied Energy & Carbon: Select materials with low energy requirements in production and low greenhouse gas emissions.



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Material Sustainability in Engineering

- Examples in Engineering
- Construction Engineering:
 - Fly ash, slag, silica fume as supplementary cementitious materials.
 - Bamboo as reinforcement in low-cost housing.
- Mechanical/Automotive Engineering:
 - Lightweight aluminum and composites to improve fuel efficiency.
- Electronics:
 - Eco-design with recyclable plastics and reduced rare earth metal use.



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7 Key Sustainable Design Principles

1. Systems Thinking

- Look at the big picture: how the design interacts with environment, society, and economy.

2. Life Cycle Approach

- Consider the full journey of a product/building: extraction → production → use → reuse/recycle.

3. Resource & Energy Efficiency

- Use less material, water, and energy while maximizing performance.

4. Use of Green & Circular Materials

- Choose renewable, recycled, and low-carbon materials that can be reused or recycled.



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7 Key Sustainable Design Principles

5. Durability & Flexibility

- Design for long life, easy repair, adaptability, and future use changes.

6. Minimize Environmental & Carbon Impact

- Reduce emissions, waste, and pollution; integrate renewable energy where possible.

7. Human & Social Well-Being

- Ensure safety, health, inclusivity, and benefits for communities and users.



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